C15 Alcohols – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 1: Production and Oxidation **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Equation for fermentation of glucose				$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$
2	Conditions for fermentation				Yeast, about 35 °C, anaerobic
3	Catalyst for hydration of ethene				H_3PO_4
4	Biofuel				Fuel made from biological material
5	Why bioethanol is claimed to be carbon neutral				CO_2 released = CO_2 absorbed by the crop
6	Oxidising agent for alcohols				Acidified potassium dichromate(VI)
7	Colour change of acidified dichromate				Orange to green
8	Oxidation products of a primary alcohol				Aldehyde, then carboxylic acid
9	Oxidation product of a secondary alcohol				Ketone
10	Tertiary alcohol with acidified dichromate				No reaction

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA A-level Chemistry • Set 2: Conditions, Products and Dehydration **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Condition to obtain an aldehyde				Distil off as it forms
2	Condition to obtain a carboxylic acid				Heat under reflux, excess oxidising agent
3	Symbol for the oxidising agent in equations				[O]
4	Product of ethanol oxidised with distillation				Ethanal
5	Product of ethanol oxidised under reflux				Ethanoic acid
6	Oxidation product of propan-2-ol				Propanone
7	Catalyst for dehydration of alcohols				Conc H_2SO_4 or H_3PO_4
8	Products of dehydration of an alcohol				Alkene + water
9	Number of alkenes from dehydrating butan-2-ol				3
10	Benefit of making alkenes from alcohols				Polymers without using crude oil

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA A-level Chemistry • Set 3: Tests **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Tollens' reagent				$[Ag(NH_3)_2]^+$
2	Positive result with Tollens'				Silver mirror
3	Positive result with Fehling's				Blue solution to brick-red precipitate
4	Active ion in Fehling's solution				Cu^{2+}
5	Ketones with Tollens' or Fehling's				No change
6	Test for a carboxylic acid				$NaHCO_3$
7	Positive result for a carboxylic acid				Effervescence
8	Gas given off by a carboxylic acid with $NaHCO_3$				CO_2
9	Primary alcohol				–OH carbon bonded to one other carbon
10	Why reflux is used				Heats without losing volatile substances

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.